

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 2, all 24 PYQ constructions, 2069–2082 BS

2. RF and M/W Transmission Lines: Numerical Problems

24 solved PYQ constructions · **≈230 marks** across 24 papers · 1 chart reading, 6 single-stub, 17 double-stub

READ THIS FIRST

Every problem below is worked as a **Smith chart construction**, step by step — which circle to draw, which arc to follow, what to read off the rim — because that is what the examiner marks. The numbers were computed exactly from the transmission-line equations, so they are what a perfectly drawn chart would give; your hand-drawn readings will differ in the second decimal place and that is fine.

- **Two solutions are normal.** Single-stub and double-stub tuners each have two valid answers. Quote either — but say which one you chose, and why.
- **Always check the forbidden region** before starting a double stub: $g_1 \leq 1/\sin^2 \beta d$. Stating this earns marks even when the load passes.
- **Never drop the S-matrix rider** — §2.8, two lines, 2–6 marks.

M. The Three Methods, in Brief

M1 — Reading the chart (Problem 1)

1. Normalise $z_L = Z_L/Z_0$ and plot.
2. Swing the **SWR circle** about the centre through that point. $|\Gamma_L|$ is its radius; S is where it cuts the right-hand real axis.
3. Moving along the line is **rotation at constant radius**: clockwise toward the generator, one lap = $\lambda/2$. Read distances on the WTG rim scale.
4. The circle crosses the horizontal axis twice per lap — those are the only **purely resistive** points, at $V_{\max}$ ($r = S$) and $V_{\min}$ ($r = 1/S$).

M2 — Single-stub shunt tuner (Problems 2–7)

1. Plot z_L , half-turn to y_L (shunt stubs *add susceptance*, so all stub work is done in admittance).
2. Rotate clockwise to the $g = 1$ **circle**. It is cut **twice** → d_1, d_2 .
3. There $y = 1 \pm jb$. The stub must supply $\mp jb$.
4. For the length, start at the stub's own termination — **open** stub at $y = 0$, **short** stub at $y = \infty$ — and run clockwise round the rim to that susceptance.
 $\ell_{\text{short}} = \ell_{\text{open}} \pm \lambda/4$.

M3 — Double-stub shunt tuner (Problems 8–24)

1. Plot z_L , half-turn to y_L ; if the first stub is offset, rotate clockwise to get y_1 , the admittance **just before stub 1**.
2. **Check** $g_1 \leq 1/\sin^2 \beta d$. If it fails, no length of stub can match this load at this spacing.
3. Draw the **spacing circle**: the $g = 1$ circle rotated the stub spacing d *toward the load*.
4. Stub 1 can only slide the point **along its own constant- g circle**. Follow that circle to its two intersections with the spacing circle → y_a, y'_a . Then $b_1 = \text{Im}(y_a) - \text{Im}(y_1)$.
5. Rotate y_a clockwise by d on a new SWR circle. It lands on $g = 1$ at $y_2 = 1 + jb$.
6. Stub 2 supplies $-jb$. Read both lengths off the rim as in M2.

Sign conventions that decide the answer

- **Clockwise = toward the generator.** One full lap is $\lambda/2$, so a quarter turn is $\lambda/8$ and $3\lambda/8$ is three quarter-turns.
- On the **admittance** chart the terminations swap ends: $y = 0$ (open) is at the left, $y = \infty$ (short) at the right — the mirror of the impedance chart. Getting this backwards corrupts every stub length by $\lambda/4$.
- Upper half = $+b$ = **capacitive**; lower half = $-b$ = inductive.

Analytic back-check

$$B = \frac{1 \pm \sqrt{g_1(1+t^2) - g_1^2 t^2}}{t}, \quad t = \tan \beta d, \quad b_1 = B - b_L$$

Note the leading 1 in the numerator, not g_1 .

Problem 1 — Smith chart reading of a mismatched line 10

79 Ch · $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 79 Ch [10]

A lossless $50\ \Omega$ line is terminated by $75 + j100\ \Omega$. Using the Smith chart find (a) Γ_L , (b) VSWR, (c) Z_{in} at 0.375λ from the load, (d) the shortest length of line for which the impedance is purely resistive, and (e) the value of that resistance.

Given

– $Z_L = 75.00 + j100.00\ \Omega$, $Z_0 = 50\ \Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 75.00 + j100.00$ over $50 = 1.50 + j2.00$.
- Plot it where the $r = 1.50$ circle cuts the $x = +2.00$ arc; it reads 0.198λ on the WTG scale.

Step 2 — draw the SWR circle and read (a) and (b)

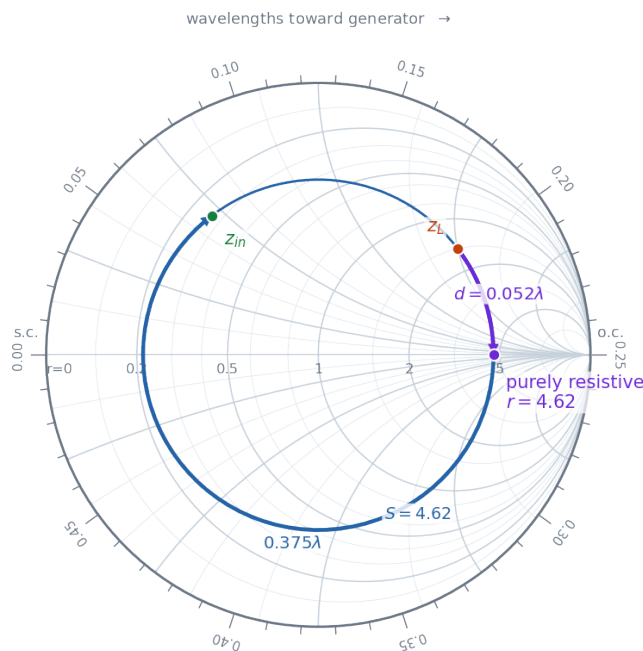
- Compass on chart centre, swing a circle through z_L : the SWR circle. Lay its radius (dividers) on the radially scaled scales below the chart.
- (a) Refl. coeff. E or I scale: $|\Gamma_L| = 0.644$; straight edge through z_L to the angle of refl. coeff. scale: 37.3° . $\Gamma_L = 0.644\angle 37.3^\circ$.
- (b) SWR scale: $S = 4.62$, also the r where the circle cuts the right-hand real axis. Check: $(z_L - 1)/(z_L + 1) = 0.644\angle 37.3^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.62$.

Step 3 — rotate 0.375λ toward the generator for (c)

- z_L reads 0.198λ WTG. Add 0.375λ : $0.198 + 0.375 = 0.573$, and since this exceeds 0.5 , subtract half a turn $\rightarrow 0.073\lambda$.
- Staying on the SWR circle (the radius never changes on a lossless line), read $z_{in} = 0.27 + j0.47$.
- (c) $Z_{in} = Z_0 z_{in} = 50 \times (0.27 + j0.47) = 13.33 + j23.33\ \Omega$.

Step 4 — shortest purely resistive point, (d) and (e)

- z is purely resistive wherever the SWR circle crosses the **horizontal axis** ($x = 0$) — which happens twice per lap, at the voltage maximum (right) and the voltage minimum (left).
- Rotating clockwise from z_L at 0.198λ , the nearer crossing is the **voltage maximum**.
- (d) $d = 0.052\lambda$.
- (e) At a voltage maximum the normalised resistance is $r = S = 4.62$, so $R = 50 \times 4.617 = 230.8\ \Omega$.



Problem 2 — Single-stub shunt tuner 2 8

72 Ash · $Z_0 = 50 \Omega$

QUESTION AS PRINTED 72 Ash [2+8]

Design single-stub shunt tuning networks (short-circuited and open-circuited) for $Z_L = 80 + j100 \Omega$, and sketch the realisation in microstrip.

Given

- $Z_L = 80.00 + j100.00 \Omega$, $Z_0 = 50 \Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 80.00 + j100.00$ over $50 = 1.60 + j2.00$.
- Plot it where the $r = 1.60$ circle cuts the $x = +2.00$ arc; it reads 0.200λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.637$ on *refl. coeff. E or I*, $S = 4.50$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.637 \angle 35.7^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.50$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.24 - j0.30$, at 0.450λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.637$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j1.65$ at 0.180λ WTG, so $d_1 = (0.500 - 0.450) + 0.180 = 0.230\lambda$.
 - $y_2 = 1.00 - j1.65$ at 0.320λ WTG, so $d_2 = (0.500 - 0.450) + 0.320 = 0.370\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 — read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ($y = \infty$, the right of the chart):
 - needs $b = -1.651$, giving $\ell_1 = 0.087\lambda$.
 - needs $b = +1.651$, giving $\ell_2 = 0.413\lambda$.
- **Open-circuited stub** — start at the open-circuit point ($y = 0$, the left of the chart):
 - needs $b = -1.651$, giving $\ell_1 = 0.337\lambda$.
 - needs $b = +1.651$, giving $\ell_2 = 0.163\lambda$.

Answer

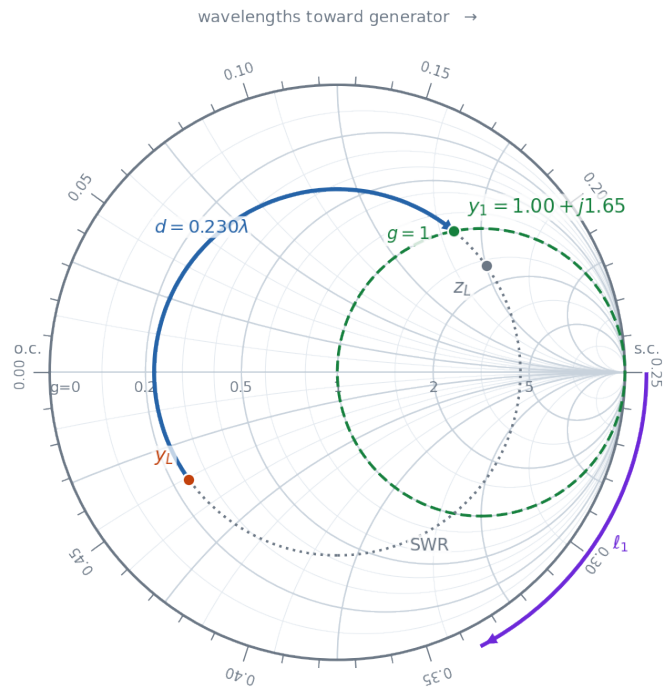
Quantity	Solution 1	Solution 2
Stub distance d	0.230λ	0.370λ
Line admittance	$1.00 + j1.65$	$1.00 - j1.65$
Short stub ℓ	0.087λ	0.413λ
Open stub ℓ	0.337λ	0.163λ

Which to build

Solution 1 puts the stub closer to the load (0.230λ against 0.370λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

Microstrip realisation

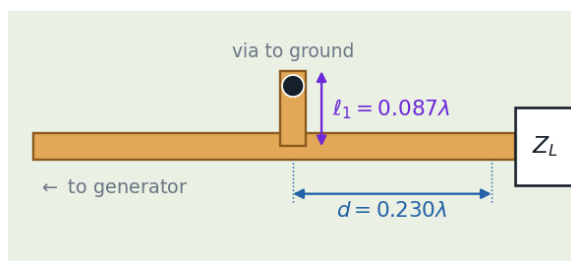
- Take FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm. Hammerstad synthesis for 50Ω gives $W/h = 1.91$, i.e. $W = 3.06$ mm, with $\epsilon_e = 3.33$.
- No frequency is given, so leave the answers in wavelengths; multiply by $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$ once f is fixed.
- Draw the stubs as T-junctions off the W -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance $\Delta\ell \approx 0.3h = 0.48$ mm.



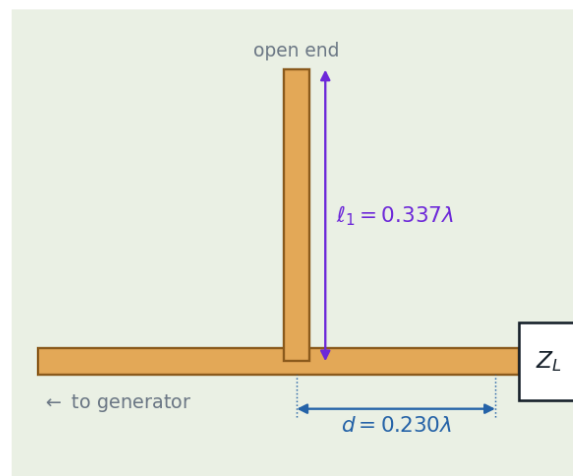
solution 1: $d = 0.230\lambda$, short stub $\ell = 0.087\lambda$

Problem 2 — Smith chart construction. Arcs are numbered in the order of the steps above.

Short-circuited stub, solution 1



Open-circuited stub, solution 1



FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, $50\ \Omega$ strip $W = 3.06$ mm, $\epsilon_e = 3.33$
 lengths in λ_g and to scale; strip width not to scale
 cut each open end $\Delta\ell = 0.48$ mm short for fringing

Problem 2 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

Problem 3 — Single-stub shunt tuner 8

74 Bh · $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 74 Bh [8]

A load presents $\Gamma_L = 0.5\angle 51^\circ$ on a $50\ \Omega$ line. Design single short-circuited and open-circuited shunt stubs using the Smith chart.

Given

$$- \Gamma_L = 0.500\angle 51.0^\circ, Z_0 = 50\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 60.42 + j62.60\ \Omega.$$

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 60.42 + j62.60$ over $50 = 1.21 + j1.25$.
- Plot it where the $r = 1.21$ circle cuts the $x = +1.25$ arc; it reads 0.179λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.500$ on *refl. coeff. E or I*, $S = 3.00$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.500\angle 51.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.00$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.40 - j0.41$, at 0.429λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.500$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j1.15$ at 0.167λ WTG, so $d_1 = (0.500 - 0.429) + 0.167 \approx 0.237\lambda$.
 - $y_2 = 1.00 - j1.15$ at 0.333λ WTG, so $d_2 = (0.500 - 0.429) + 0.333 = 0.404\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 — read each stub length off the rim

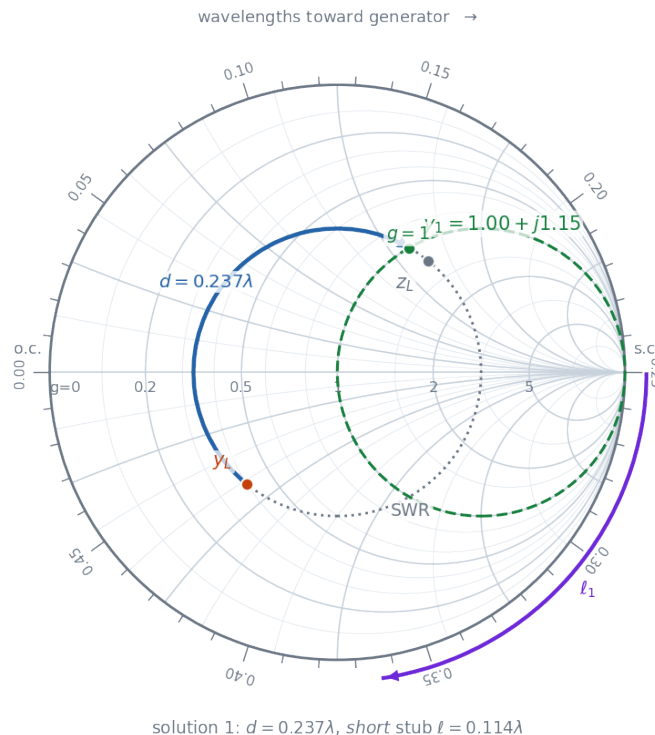
- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ($y = \infty$, the right of the chart):
 - needs $b = -1.155$, giving $\ell_1 = 0.114\lambda$.
 - needs $b = +1.155$, giving $\ell_2 = 0.386\lambda$.
- **Open-circuited stub** — start at the open-circuit point ($y = 0$, the left of the chart):
 - needs $b = -1.155$, giving $\ell_1 = 0.364\lambda$.
 - needs $b = +1.155$, giving $\ell_2 = 0.136\lambda$.

Answer

Quantity	Solution 1	Solution 2
Stub distance d	0.237λ	0.404λ
Line admittance	$1.00 + j1.15$	$1.00 - j1.15$
Short stub ℓ	0.114λ	0.386λ
Open stub ℓ	0.364λ	0.136λ

Which to build

Solution 1 puts the stub closer to the load (0.237λ against 0.404λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.



Problem 3 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 4 — Single-stub shunt tuner 5 3 2

81 Bh · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 81 Bh [5+3+2]

A $100\ \Omega$ lossless line feeds a complex load $120 - j160\ \Omega$. Give all design steps of a single matching stub with reasoning, and the S-matrix before and after matching.

Given

- $Z_L = 120.00 - j160.00\ \Omega$, $Z_0 = 100\ \Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 120.00 - j160.00$ over $100 = 1.20 - j1.60$.
- Plot it where the $r = 1.20$ circle cuts the $x = -1.60$ arc; it reads 0.315λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.593$ on refl. coeff. E or I , $S = 3.91$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.593\angle -46.8^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.91$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the admittance grid in place and skip this step.
- $y_L = 1/z_L = 0.30 + j0.40$, at 0.065λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.593$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j1.47$ at 0.175λ WTG, so $d_1 = 0.175 - 0.065 = 0.110\lambda$.
 - $y_2 = 1.00 - j1.47$ at 0.325λ WTG, so $d_2 = 0.325 - 0.065 \approx 0.259\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 — read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.

– **Short-circuited stub** – start at the short-circuit point ($y = \infty$, the right of the chart):

- needs $b = -1.472$, giving $\ell_1 = 0.095\lambda$.
- needs $b = +1.472$, giving $\ell_2 = 0.405\lambda$.

Answer

Quantity	Solution 1	Solution 2
Stub distance d	0.110λ	0.259λ
Line admittance	$1.00 + j1.47$	$1.00 - j1.47$
Short stub ℓ	0.095λ	0.405λ

Which to build

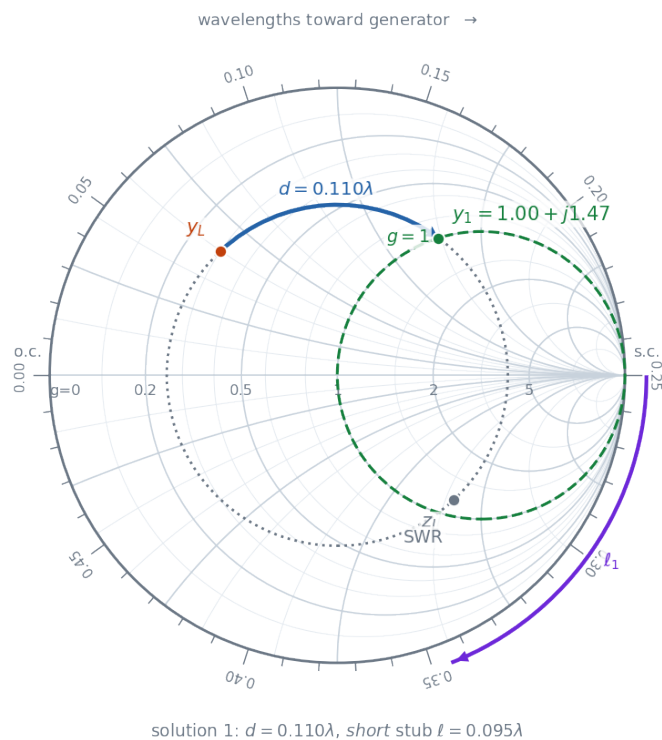
Solution 1 puts the stub closer to the load (0.110λ against 0.259λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.593\angle -46.8^\circ]$, with $S = 3.91$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 4 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 5 – Single-stub shunt tuner 10

79 Ch · $Z_0 = 50\Omega$

QUESTION AS PRINTED 79 Ch [10]

Design a single short-circuited matching stub with self-defined line and load impedances, mentioning the steps.

Given

- $Z_L = 35.00 - j60.00\Omega$, $Z_0 = 50\Omega$.

- **verified/added** The paper leaves the values to the candidate; $Z_L = 35.00 - j60.00 \Omega$ on a 50Ω line is assumed here. Any complex load works – the method is what earns the marks.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 35.00 - j60.00$ over $50 = 0.70 - j1.20$.
- Plot it where the $r = 0.70$ circle cuts the $x = -1.20$ arc; it reads 0.346λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.594$ on *refl. coeff.* E or I , $S = 3.93$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.594 \angle -68.8^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.93$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.36 + j0.62$, at 0.096λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.594$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j1.48$ at 0.176λ WTG, so $d_1 = 0.176 - 0.096 = 0.080\lambda$.
 - $y_2 = 1.00 - j1.48$ at 0.324λ WTG, so $d_2 = 0.324 - 0.096 \approx 0.229\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 – read each stub length off the rim

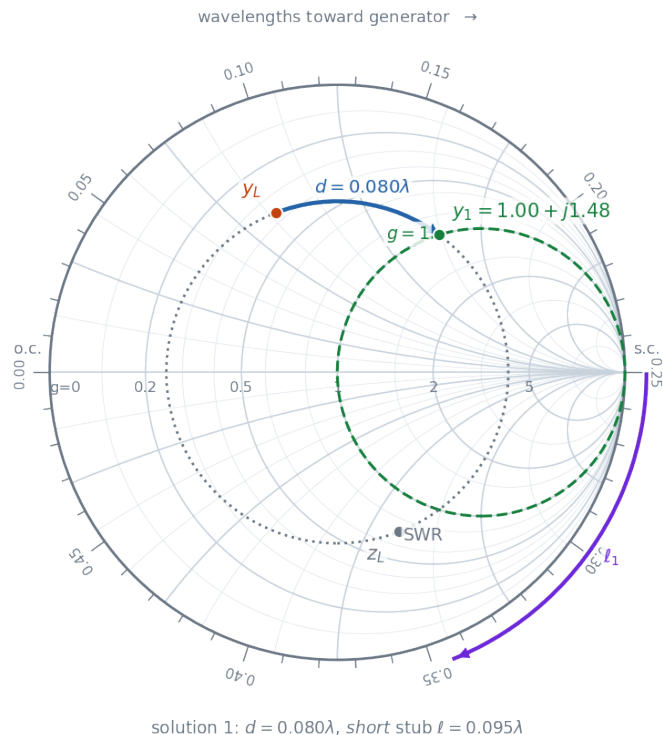
- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** – start at the short-circuit point ($y = \infty$, the right of the chart):
 - needs $b = -1.478$, giving $\ell_1 = 0.095\lambda$.
 - needs $b = +1.478$, giving $\ell_2 = 0.405\lambda$.

Answer

Quantity	Solution 1	Solution 2
Stub distance d	0.080λ	0.229λ
Line admittance	$1.00 + j1.48$	$1.00 - j1.48$
Short stub ℓ	0.095λ	0.405λ

Which to build

Solution 1 puts the stub closer to the load (0.080λ against 0.229λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.



Problem 5 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 6 — Single-stub shunt tuner 10

76 Bh · $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 76 Bh [10]

Design a single-stub tuner matching a lossless line to an antenna load (any assumed placement and length) and derive its S-matrix.

Given

- $Z_L = 73.00 + j42.50\ \Omega$, $Z_0 = 50\ \Omega$.
- verified/added The paper leaves the values to the candidate; $Z_L = 73.00 + j42.50\ \Omega$ on a $50\ \Omega$ line is assumed here. Any complex load works — the method is what earns the marks.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 73.00 + j42.50$ over $50 = 1.46 + j0.85$.
- Plot it where the $r = 1.46$ circle cuts the $x = +0.85$ arc; it reads 0.191λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.371$ on refl. coeff. E or I , $S = 2.18$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.371\angle 42.5^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.18$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.51 - j0.30$, at 0.441λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.371$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j0.80$ at 0.155λ WTG, so $d_1 = (0.500 - 0.441) + 0.155 = 0.214\lambda$.
 - $y_2 = 1.00 - j0.80$ at 0.345λ WTG, so $d_2 = (0.500 - 0.441) + 0.345 = 0.404\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 — read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ($y = \infty$, the right of the chart):
 - needs $b = -0.800$, giving $\ell_1 = 0.143\lambda$.
 - needs $b = +0.800$, giving $\ell_2 = 0.357\lambda$.

Answer

Quantity	Solution 1	Solution 2
Stub distance d	0.214λ	0.404λ
Line admittance	$1.00 + j0.80$	$1.00 - j0.80$
Short stub ℓ	0.143λ	0.357λ

Which to build

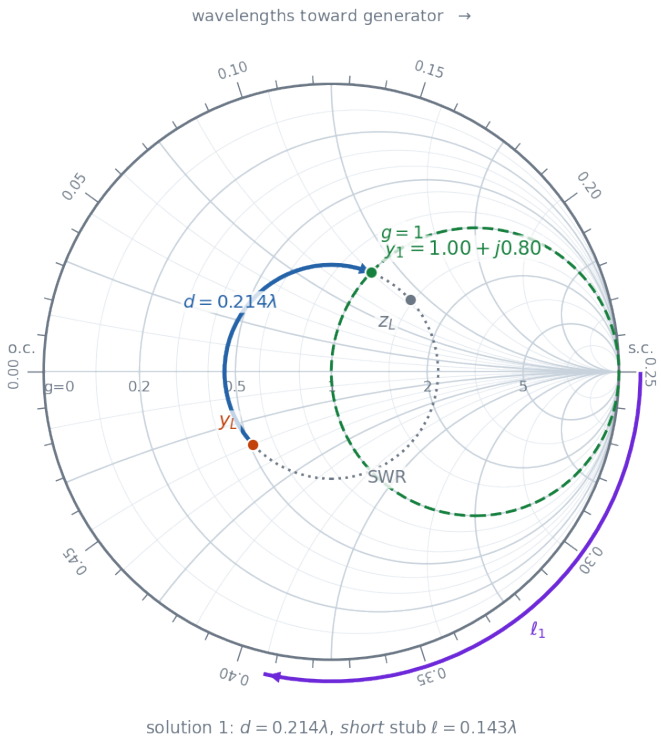
Solution 1 puts the stub closer to the load (0.214λ against 0.404λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

S-matrix before and after matching

- **Before** — the bare load is a one-port, so $[S] = [\Gamma_L] = [0.371\angle 42.5^\circ]$, with $S = 2.18$.
- **After** — the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 6 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 7 – Single-stub shunt tuner 4 4 2

72 Ma · $Z_0 = 75 \Omega$

QUESTION AS PRINTED 72 Ma [4+4+2]

Design shunt short- and open-stub matching networks using the Smith chart for $Z_0 = 75 \Omega$, $Z_L = 78.27 + j60.93 \Omega$, and sketch the physical diagram considering microstrips.

Given

- $Z_L = 78.27 + j60.93 \Omega$, $Z_0 = 75 \Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 78.27 + j60.93$ over $75 = 1.04 + j0.81$.
- Plot it where the $r = 1.04$ circle cuts the $x = +0.81$ arc; it reads 0.159λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.370$ on refl. coeff. E or I , $S = 2.17$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.370 \angle 65.2^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.17$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.60 - j0.46$, at 0.409λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – walk to the $g = 1$ circle

- Swing the SWR circle (radius $|\Gamma_L| = 0.370$) through y_L .
- Rotate **clockwise** (toward the generator) to each intersection with the $g = 1$ circle. There are always two:
 - $y_1 = 1.00 + j0.80$ at 0.155λ WTG, so $d_1 = (0.500 - 0.409) + 0.155 = 0.246\lambda$.
 - $y_2 = 1.00 - j0.80$ at 0.345λ WTG, so $d_2 = (0.500 - 0.409) + 0.345 \approx 0.435\lambda$.
- Only the *susceptance* is left to cancel: the conductance is already 1.

Step 4 – read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** – start at the short-circuit point ($y = \infty$, the right of the chart):
 - needs $b = -0.796$, giving $\ell_1 = 0.143\lambda$.
 - needs $b = +0.796$, giving $\ell_2 = 0.357\lambda$.
- **Open-circuited stub** – start at the open-circuit point ($y = 0$, the left of the chart):
 - needs $b = -0.796$, giving $\ell_1 = 0.393\lambda$.
 - needs $b = +0.796$, giving $\ell_2 = 0.107\lambda$.

Answer

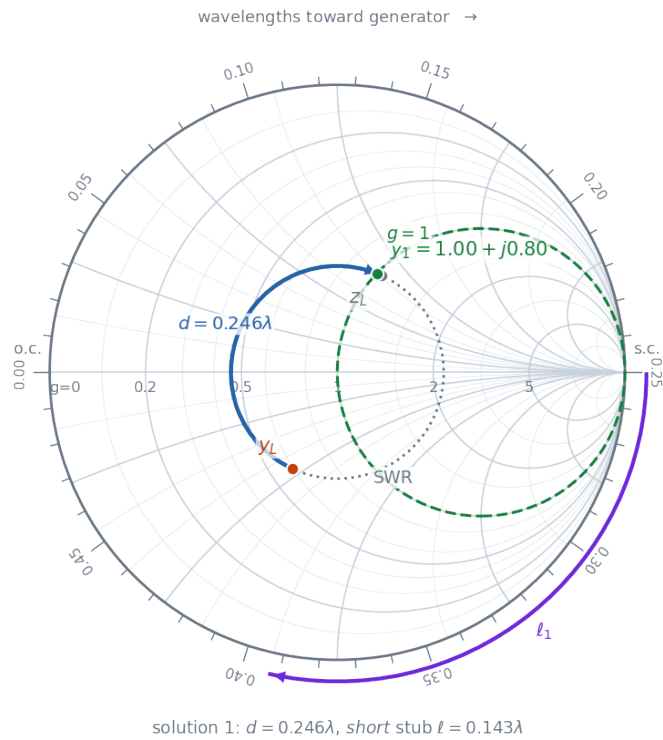
Quantity	Solution 1	Solution 2
Stub distance d	0.246λ	0.435λ
Line admittance	$1.00 + j0.80$	$1.00 - j0.80$
Short stub ℓ	0.143λ	0.357λ
Open stub ℓ	0.393λ	0.107λ

Which to build

Solution 1 puts the stub closer to the load (0.246λ against 0.435λ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

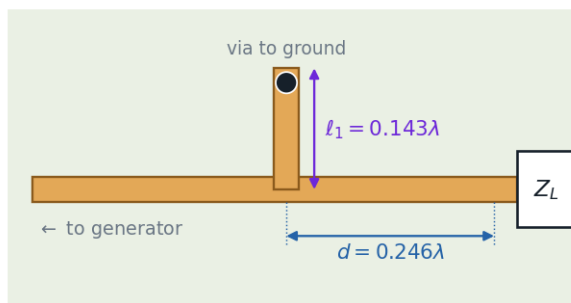
Microstrip realisation

- Take FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm. Hammerstad synthesis for 75Ω gives $W/h = 0.89$, i.e. $W = 1.43$ mm, with $\epsilon_e = 3.15$.
- No frequency is given, so leave the answers in wavelengths; multiply by $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$ once f is fixed.
- Draw the stubs as T-junctions off the W -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance $\Delta\ell \approx 0.3h = 0.48$ mm.

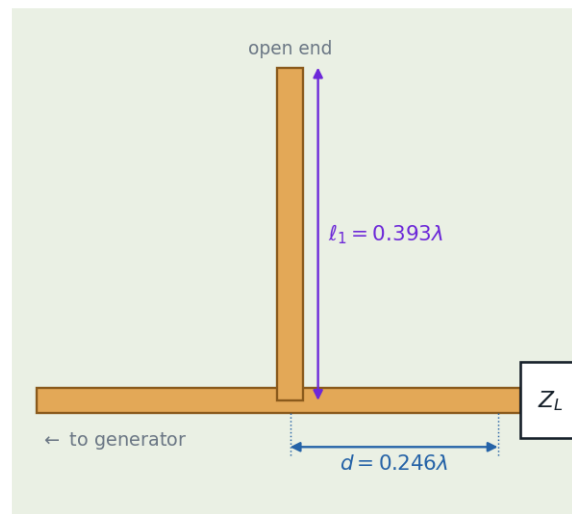


Problem 7 — Smith chart construction. Arcs are numbered in the order of the steps above.

Short-circuited stub, solution 1



Open-circuited stub, solution 1



FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, 75Ω strip $W = 1.43$ mm, $\epsilon_e = 3.15$
 lengths in λ_g and to scale; strip width not to scale
 cut each open end $\Delta l = 0.48$ mm short for fringing

Problem 7 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

Problem 8 — Double-stub shunt tuner 10

71 Bh · $Z_0 = 300\ \Omega$

QUESTION AS PRINTED 71 Bh [10]

Design a double-stub tuner ($3\lambda/8$ spacing) for $Z_L = 300 + j300\ \Omega$ on a $300\ \Omega$ line. Include the figure.

Given

– $Z_L = 300.00 + j300.00\ \Omega$, $Z_0 = 300\ \Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 300.00 + j300.00$ over $300 = 1.00 + j1.00$.
- Plot it where the $r = 1.00$ circle cuts the $x = +1.00$ arc; it reads 0.162λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.447$ on *refl. coeff. E or I*, $S = 2.62$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.447\angle 63.4^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.62$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.50 - j0.50$, at 0.412λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.50 - j0.50$.

Step 4 — draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.500$. ✓

Step 5 — stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.500$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.50 - j1.87$, so stub 1 must add $b_1 = -1.866 - (-0.500) = -1.366$.
 - **Solution B:** $y'_a = 0.50 - j0.13$, so $b'_1 = +0.366$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.101\lambda$ (A) and 0.306λ (B).

Step 6 — the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.73$ (A), $y'_2 = 1.00 - j0.73$ (B).

Step 7 — stub 2 cancels what is left

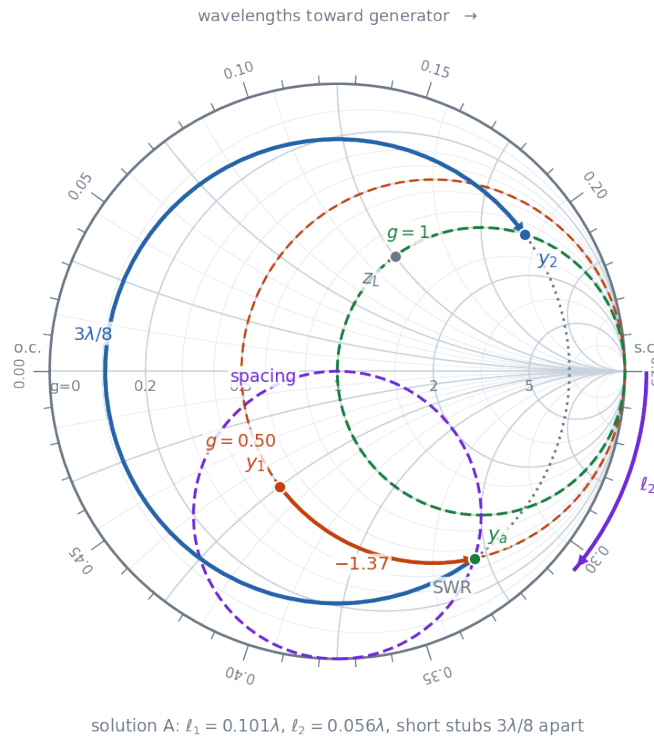
- Stub 2 supplies $b_2 = -2.732$ (A) or $+0.732$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 300\ \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.056\lambda$ (A) and 0.351λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.50 - j0.50$	$0.50 - j0.50$
Stub 1 susceptance b_1	-1.366	$+0.366$
Stub 1 length ℓ_1	0.101λ	0.306λ
y after the spacing	$1.00 + j2.73$	$1.00 - j0.73$
Stub 2 susceptance b_2	-2.732	$+0.732$
Stub 2 length ℓ_2	0.056λ	0.351λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 8 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 9 — Double-stub shunt tuner 10

70 Bh · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 70 Bh [10]

Design a double-stub tuner ($3\lambda/8$ spacing) for $Z_L = 80 + j180\ \Omega$ on a $100\ \Omega$ line at 3 GHz, with figure.

Given

- $Z_L = 80.00 + j180.00\ \Omega$, $Z_0 = 100\ \Omega$.
- $f = 3\ \text{GHz} \Rightarrow \lambda_0 = c/f = 99.9\ \text{mm}$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 80.00 + j180.00$ over 100 = $0.80 + j1.80$.
- Plot it where the $r = 0.80$ circle cuts the $x = +1.80$ arc; it reads 0.179λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.711$ on *refl. coeff.* *E or I*, $S = 5.93$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.711\angle 51.3^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 5.93$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.21 - j0.46$, at 0.429λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.21 - j0.46$.

Step 4 — draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.206$. ✓

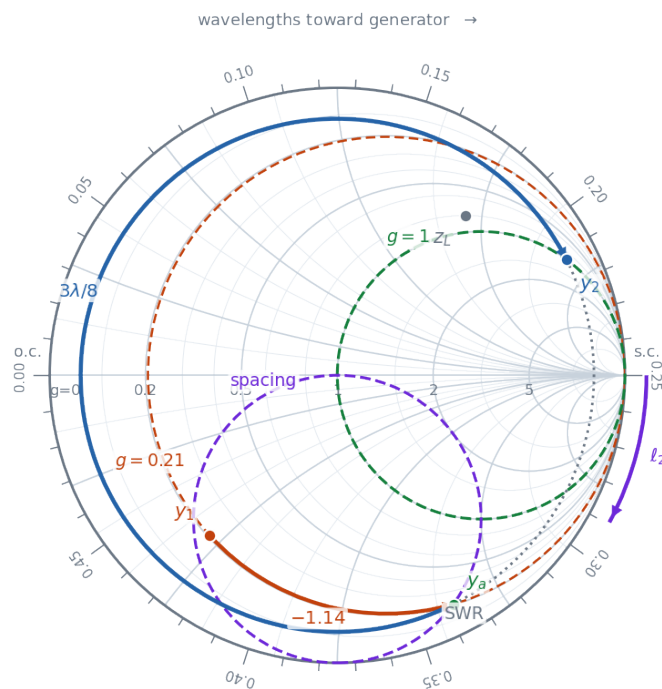
- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.206$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.21 - j1.61$, so stub 1 must add $b_1 = -1.608 - (-0.464) = -1.144$.
 - **Solution B:** $y'_a = 0.21 - j0.39$, so $b'_1 = +0.072$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.114\lambda$ (A) and 0.261λ (B).

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.95$ (A), $y'_2 = 1.00 - j1.95$ (B).

- Stub 2 supplies $b_2 = -3.950$ (A) or $+1.950$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 100 \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.039\lambda$ (A) and 0.425λ (B).

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.21 - j0.46$	$0.21 - j0.46$
Stub 1 susceptance	-1.144	$+0.072$
b_1		
Stub 1 length ℓ_1	0.114λ	0.261λ
y after the spacing	$1.00 + j3.95$	$1.00 - j1.95$
Stub 2 susceptance	-3.950	$+1.950$
b_2		
Stub 2 length ℓ_2	0.039λ	0.425λ

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



solution A: $\ell_1 = 0.114\lambda$, $\ell_2 = 0.039\lambda$, short stubs $3\lambda/8$ apart

Problem 10 – Double-stub shunt tuner 10

70 Ma · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 70 Ma [10]

Design a double-stub tuner ($3\lambda/8$ spacing) for $Z_L = 190 + j110\ \Omega$ on a $100\ \Omega$ line at 10 GHz, with figure.

Given

- $Z_L = 190.00 + j110.00\ \Omega$, $Z_0 = 100\ \Omega$.
- $f = 10\ \text{GHz} \Rightarrow \lambda_0 = c/f = 30.0\ \text{mm}$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 190.00 + j110.00$ over 100 = 1.90 + j1.10.
- Plot it where the $r = 1.90$ circle cuts the $x = +1.10$ arc; it reads 0.208λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.458$ on *refl. coeff. E or I*, $S = 2.69$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.458\angle 29.9^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.69$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.39 - j0.23$, at 0.458λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.39 - j0.23$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.394$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.394$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.39 - j1.80$, so stub 1 must add $b_1 = -1.796 - (-0.228) = -1.567$.
 - **Solution B:** $y'_a = 0.39 - j0.20$, so $b'_1 = +0.024$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.090\lambda$ (A) and 0.254λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.02$ (A), $y'_2 = 1.00 - j1.02$ (B).

Step 7 – stub 2 cancels what is left

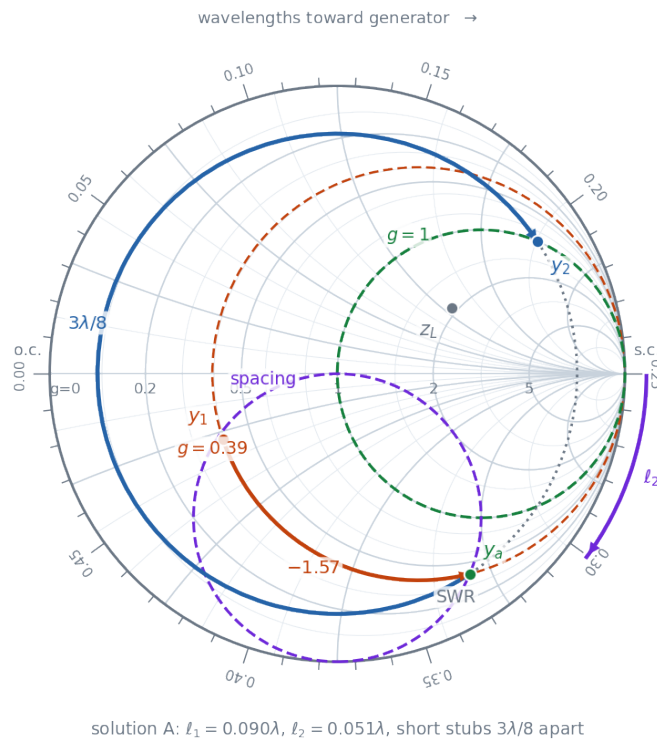
- Stub 2 supplies $b_2 = -3.018$ (A) or $+1.018$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 100\ \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.051\lambda$ (A) and 0.376λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.39 - j0.23$	$0.39 - j0.23$
Stub 1 susceptance b_1	-1.567	$+0.024$
Stub 1 length ℓ_1	0.090λ	0.254λ
y after the spacing	$1.00 + j3.02$	$1.00 - j1.02$
Stub 2 susceptance b_2	-3.018	$+1.018$
Stub 2 length ℓ_2	0.051λ	0.376λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 10 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 11 — Double-stub shunt tuner 8

73 Ma · $Z_0 = 50 \Omega$

QUESTION AS PRINTED 73 Ma [8]

Design a double-stub tuner for an inductive load on a 50Ω line. Include the figure.

Given

- $Z_L = 40.00 + j70.00 \Omega$, $Z_0 = 50 \Omega$.
- verified/added The paper leaves the values to the candidate; $Z_L = 40.00 + j70.00 \Omega$ on a 50Ω line is assumed here. Any complex load works — the method is what earns the marks.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 40.00 + j70.00$ over $50 = 0.80 + j1.40$.
- Plot it where the $r = 0.80$ circle cuts the $x = +1.40$ arc; it reads 0.166λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.620$ on refl. coeff. E or I, $S = 4.27$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.620 \angle 60.3^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.27$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.31 - j0.54$, at 0.416λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.31 - j0.54$.

Step 4 — draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ toward the load (anticlockwise). Call the result the **spacing circle**.

- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.308$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.308$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.31 - j1.72$, so stub 1 must add $b_1 = -1.722 - (-0.538) = -1.183$.
 - **Solution B:** $y'_a = 0.31 - j0.28$, so $b'_1 = +0.260$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.112\lambda$ (A) and 0.290λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.35$ (A), $y'_2 = 1.00 - j1.35$ (B).

Step 7 – stub 2 cancels what is left

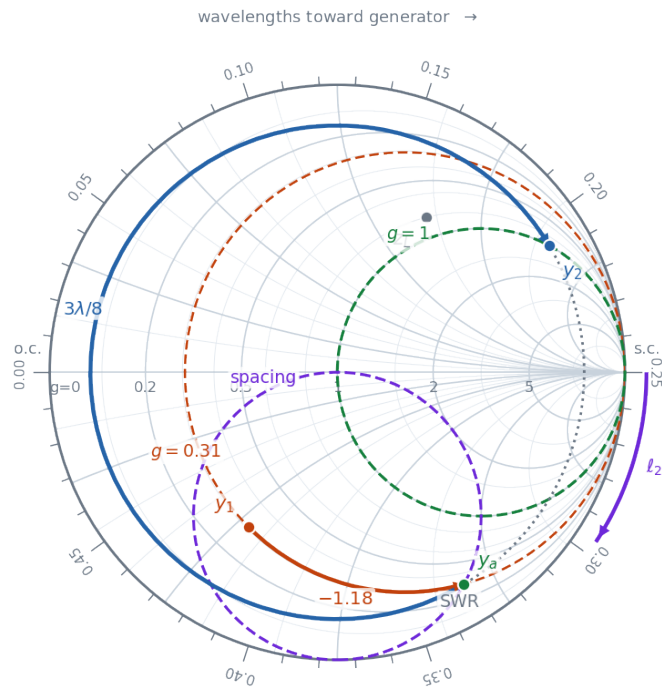
- Stub 2 supplies $b_2 = -3.345$ (A) or $+1.345$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.046\lambda$ (A) and 0.398λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.31 - j0.54$	$0.31 - j0.54$
Stub 1 susceptance b_1	-1.183	$+0.260$
Stub 1 length ℓ_1	0.112λ	0.290λ
y after the spacing	$1.00 + j3.35$	$1.00 - j1.35$
Stub 2 susceptance b_2	-3.345	$+1.345$
Stub 2 length ℓ_2	0.046λ	0.398λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



solution A: $l_1 = 0.112\lambda$, $l_2 = 0.046\lambda$, short stubs $3\lambda/8$ apart

Problem 11 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 12 – Double-stub shunt tuner 3 15

69 Bh · $Z_0 = 50\Omega$

QUESTION AS PRINTED 69 Bh [3+15]

Design a short-circuited double-stub tuner with $3\lambda/8$ spacing for $Z_L = 75 + j75\Omega$ on a 50Ω line.

Given

– $Z_L = 75.00 + j75.00\Omega$, $Z_0 = 50\Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 75.00 + j75.00$ over $50 = 1.50 + j1.50$.
- Plot it where the $r = 1.50$ circle cuts the $x = +1.50$ arc; it reads 0.194λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.542$ on *refl. coeff.* E or I, $S = 3.37$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.542\angle 40.6^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.37$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.33 - j0.33$, at 0.444λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.33 - j0.33$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.333$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.333$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:

- **Solution A:** $y_a = 0.33 - j1.75$, so stub 1 must add $b_1 = -1.745 - (-0.333) = -1.412$.
- **Solution B:** $y'_a = 0.33 - j0.25$, so $b'_1 = +0.079$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.098\lambda$ (A) and 0.262λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.24$ (A), $y'_2 = 1.00 - j1.24$ (B).

Step 7 – stub 2 cancels what is left

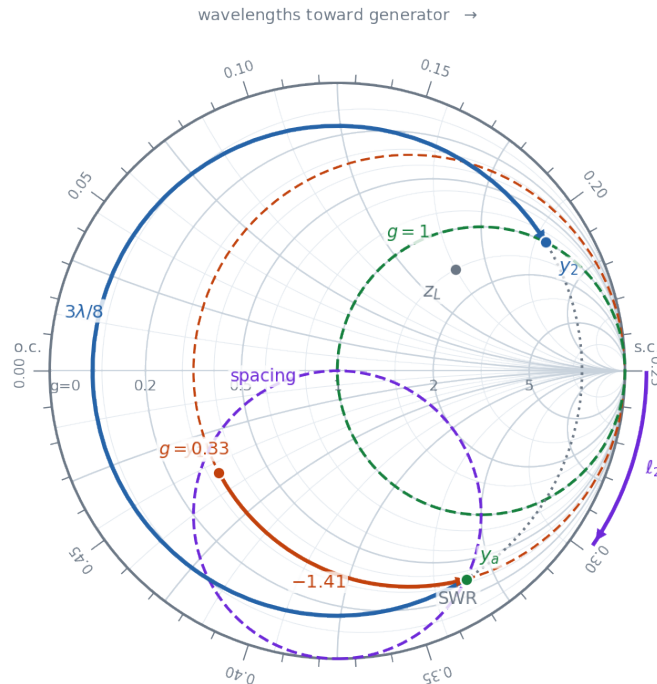
- Stub 2 supplies $b_2 = -3.236$ (A) or $+1.236$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.048\lambda$ (A) and 0.392λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.33 - j0.33$	$0.33 - j0.33$
Stub 1 susceptance	-1.412	$+0.079$
b_1		
Stub 1 length ℓ_1	0.098λ	0.262λ
y after the spacing	$1.00 + j3.24$	$1.00 - j1.24$
Stub 2 susceptance	-3.236	$+1.236$
b_2		
Stub 2 length ℓ_2	0.048λ	0.392λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



solution A: $\ell_1 = 0.098\lambda$, $\ell_2 = 0.048\lambda$, short stubs $3\lambda/8$ apart

Problem 12 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 13 – Double-stub shunt tuner 8 2

73 Bh · $Z_0 = 78 \Omega$

QUESTION AS PRINTED 73 Bh [8+2]

Design a short-circuited double-stub tuner for a complex inductive load on a 78.0Ω line, with figure.

Given

- $Z_L = 95.00 + j85.00 \Omega$, $Z_0 = 78 \Omega$.
- **verified/added** The paper leaves the values to the candidate; $Z_L = 95.00 + j85.00 \Omega$ on a 78Ω line is assumed here. Any complex load works – the method is what earns the marks.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 95.00 + j85.00$ over $78 = 1.22 + j1.09$.
- Plot it where the $r = 1.22$ circle cuts the $x = +1.09$ arc; it reads 0.177λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.450$ on *refl. coeff. E or I*, $S = 2.63$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.450 \angle 52.5^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.63$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.46 - j0.41$, at 0.427λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.46 - j0.41$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.456$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.456$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.46 - j1.84$, so stub 1 must add $b_1 = -1.839 - (-0.408) = -1.431$.
 - **Solution B:** $y'_a = 0.46 - j0.16$, so $b'_1 = +0.247$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.097\lambda$ (A) and 0.289λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.84$ (A), $y'_2 = 1.00 - j0.84$ (B).

Step 7 – stub 2 cancels what is left

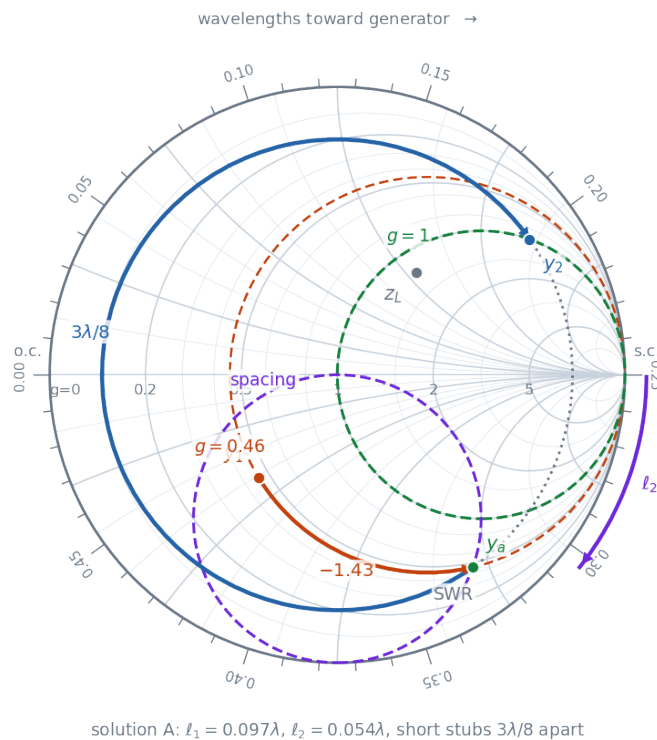
- Stub 2 supplies $b_2 = -2.840$ (A) or $+0.840$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 78 \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.054\lambda$ (A) and 0.361λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.46 - j0.41$	$0.46 - j0.41$
Stub 1 susceptance	-1.431	$+0.247$
b_1		
Stub 1 length ℓ_1	0.097λ	0.289λ
y after the spacing	$1.00 + j2.84$	$1.00 - j0.84$
Stub 2 susceptance	-2.840	$+0.840$
b_2		
Stub 2 length ℓ_2	0.054λ	0.361λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 13 — Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 14 — Double-stub shunt tuner 10 2

75 Bb · $Z_0 = 75\Omega$

QUESTION AS PRINTED 75 Bb [10+2]

A 75Ω coaxial line feeds a load of normalised impedance $z_L = 0.4 + j0.85$. Design a short-circuited double-stub tuner.

Given

– $Z_L = 30.00 + j63.75\Omega$, $Z_0 = 75\Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 30.00 + j63.75$ over $75 = 0.40 + j0.85$.
- Plot it where the $r = 0.40$ circle cuts the $x = +0.85$ arc; it reads 0.120λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.635$ on refl. coeff. E or I, $S = 4.48$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.635\angle 94.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.48$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.45 - j0.96$, at 0.370λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.45 - j0.96$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.453$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.453$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.45 - j1.84$, so stub 1 must add $b_1 = -1.837 - (-0.963) = -0.874$.
 - **Solution B:** $y'_a = 0.45 - j0.16$, so $b'_1 = +0.800$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.136\lambda$ (A) and 0.357λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.85$ (A), $y'_2 = 1.00 - j0.85$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.847$ (A) or $+0.847$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 75\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.054\lambda$ (A) and 0.362λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.45 - j0.96$	$0.45 - j0.96$
Stub 1 susceptance	-0.874	$+0.800$
b_1		
Stub 1 length ℓ_1	0.136λ	0.357λ
y after the spacing	$1.00 + j2.85$	$1.00 - j0.85$
Stub 2 susceptance	-2.847	$+0.847$
b_2		
Stub 2 length ℓ_2	0.054λ	0.362λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 15 – Double-stub shunt tuner 8 2

Step 5 – stub 1: slide along the constant- q circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.268$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.27 - j1.68$, so stub 1 must add $b_1 = -1.681 - (-0.208) = -1.473$.
 - **Solution B:** $y'_a = 0.27 - j0.32$, so $b'_1 = -0.111$.
- Read ℓ_1 off the rim, starting at the open-circuit point ($y = 0$, the left of the chart): $\ell_1 = 0.345\lambda$ (A) and 0.482λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.54$ (A), $y'_2 = 1.00 - j1.54$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -3.545$ (A) or $+1.545$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the open-circuit point ($y = 0$, the left of the chart): $\ell_2 = 0.294\lambda$ (A) and 0.159λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.400λ	0.400λ
y before stub 1	$0.27 - j0.21$	$0.27 - j0.21$
Stub 1 susceptance	-1.473	-0.111
b_1		
Stub 1 length ℓ_1	0.345λ	0.482λ
y after the spacing	$1.00 + j3.54$	$1.00 - j1.54$
Stub 2 susceptance	-3.545	$+1.545$
b_2		
Stub 2 length ℓ_2	0.294λ	0.159λ

Both are correct.

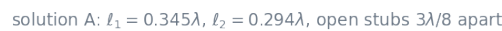
Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.593\angle -46.8^\circ]$, with $S = 3.91$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 16 – Double-stub shunt tuner 8 2

QUESTION AS PRINTED 80 Bh [8+2]

Given

- $Z_L = 109.00 + j120.00 \Omega$, $Z_0 = 75 \Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 109.00 + j120.00$ over $75 = 1.45 + j1.60$.
- Plot it where the $r = 1.45$ circle cuts the $x = +1.60$ arc; it reads 0.193λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.568$ on *refl. coeff.* *E* or *I*, $S = 3.63$ on *SWR*. **Check:** $(z_L - 1)/(z_L + 1) = 0.568 \angle 41.1^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.63$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.31 - j0.34$, at 0.443λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.31 - j0.34$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $q_1 \leq 1/\sin^2 \beta d = 2.00$, and here $q_1 = 0.311$. ✓

Step 5 – stub 1: slide along the constant- q circle

- A shunt stub adds susceptance only, so the point can move *only along its own $g = 0.311$ circle*.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:

- **Solution A:** $y_a = 0.31 - j1.72$, so stub 1 must add $b_1 = -1.725 - (-0.342) = -1.382$.
- **Solution B:** $y'_a = 0.31 - j0.28$, so $b'_1 = +0.067$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.100\lambda$ (A) and 0.261λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.33$ (A), $y'_2 = 1.00 - j1.33$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -3.330$ (A) or $+1.330$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 75\ \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.046\lambda$ (A) and 0.397λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.31 - j0.34$	$0.31 - j0.34$
Stub 1 susceptance b_1	-1.382	$+0.067$
Stub 1 length ℓ_1	0.100λ	0.261λ
y after the spacing	$1.00 + j3.33$	$1.00 - j1.33$
Stub 2 susceptance b_2	-3.330	$+1.330$
Stub 2 length ℓ_2	0.046λ	0.397λ

Both are correct.

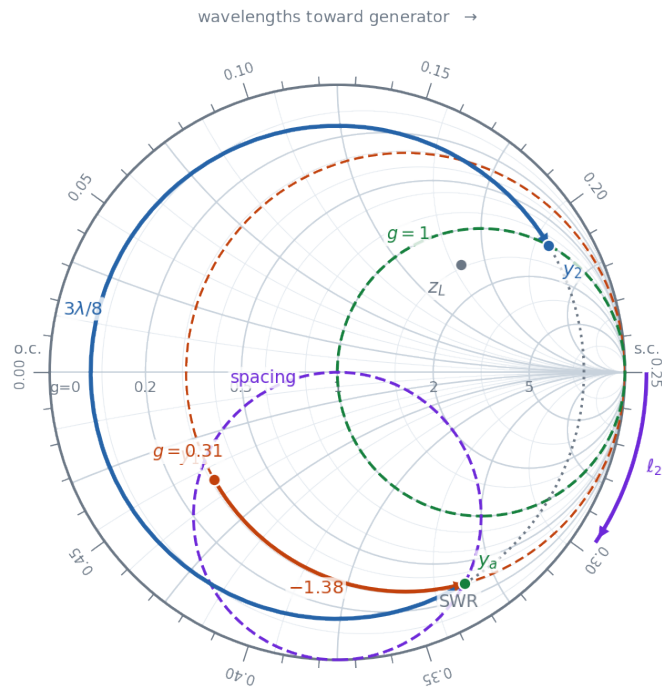
Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.568\angle 41.1^\circ]$, with $S = 3.63$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



solution A: $l_1 = 0.100\lambda$, $l_2 = 0.046\lambda$, short stubs $3\lambda/8$ apart

Problem 16 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 17 – Double-stub shunt tuner 8 2

81 Ba · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 81 Ba [8+2]

A load presents $\Gamma_L = 0.64\angle 58^\circ$ on a $100\ \Omega$ line. Design a double-stub tuner and prepare the S-matrix of the matched network.

Given

– $\Gamma_L = 0.640\angle 58.0^\circ$, $Z_0 = 100\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 80.73 + j148.43\ \Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 80.73 + j148.43$ over 100 = $0.81 + j1.48$.
- Plot it where the $r = 0.81$ circle cuts the $x = +1.48$ arc; it reads 0.169λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.640$ on *refl. coeff.* E or I, $S = 4.56$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.640\angle 58.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.56$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.28 - j0.52$, at 0.419λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.28 - j0.52$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.283$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.283$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:

- **Solution A:** $y_a = 0.28 - j1.70$, so stub 1 must add $b_1 = -1.697 - (-0.520) = -1.177$.
- **Solution B:** $y'_a = 0.28 - j0.30$, so $b'_1 = +0.217$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.112\lambda$ (A) and 0.284λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.46$ (A), $y'_2 = 1.00 - j1.46$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -3.464$ (A) or $+1.464$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 100\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.045\lambda$ (A) and 0.405λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.28 - j0.52$	$0.28 - j0.52$
Stub 1 susceptance b_1	-1.177	$+0.217$
Stub 1 length ℓ_1	0.112λ	0.284λ
y after the spacing	$1.00 + j3.46$	$1.00 - j1.46$
Stub 2 susceptance b_2	-3.464	$+1.464$
Stub 2 length ℓ_2	0.045λ	0.405λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.640\angle 58.0^\circ]$, with $S = 4.56$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 18 – Double-stub shunt tuner 8 2

QUESTION AS PRINTED 79 Bh [8+2]

Given

Step 1 – normalise and plot z_L

- ### Step 2 – convert to admittance

- ### Step 3 – walk to the first stub

- ### Step 4 – draw the spacing circle

- Step 5 – stub 1: slide along the constant- q circle**

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.425$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.43 - j1.82$, so stub 1 must add $b_1 = -1.818 - (0.385) = -2.203$.
 - **Solution B:** $y'_a = 0.43 - j0.18$, so $b'_1 = -0.567$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.068\lambda$ (A) and 0.168λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.92$ (A), $y'_2 = 1.00 - j0.92$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.925$ (A) or $+0.925$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.052\lambda$ (A) and 0.369λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.010λ	0.010λ
y before stub 1	$0.43 + j0.38$	$0.43 + j0.38$
Stub 1 susceptance	-2.203	-0.567
b_1		
Stub 1 length ℓ_1	0.068λ	0.168λ
y after the spacing	$1.00 + j2.92$	$1.00 - j0.92$
Stub 2 susceptance	-2.925	$+0.925$
b_2		
Stub 2 length ℓ_2	0.052λ	0.369λ

Both are correct.

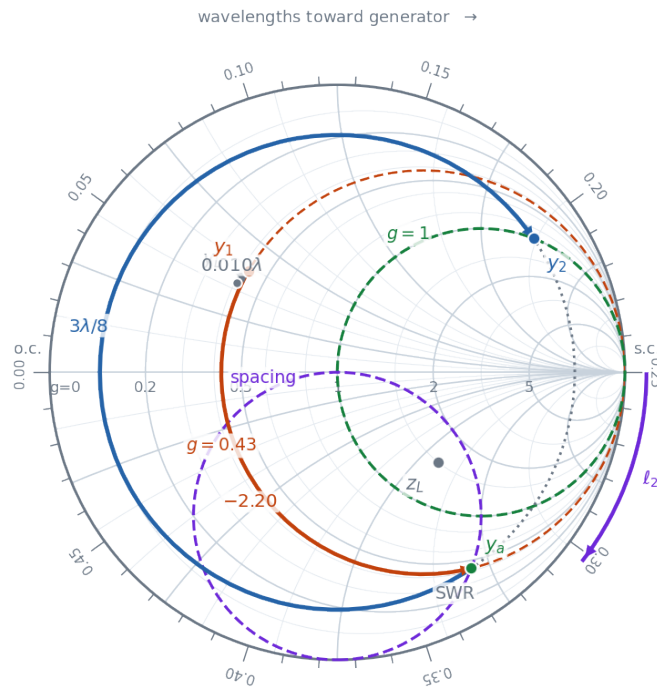
Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.469\angle -41.7^\circ]$, with $S = 2.76$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.



solution A: $l_1 = 0.068\lambda$, $l_2 = 0.052\lambda$, short stubs $3\lambda/8$ apart

Problem 18 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 19 – Double-stub shunt tuner 10

82 Bh · $Z_0 = 75\ \Omega$

QUESTION AS PRINTED 82 Bh [10]

A broadband microstrip antenna has load reflection coefficient $0.33\angle 66^\circ$ on a $75\ \Omega$ transmission patch. Design the matching stubs and express the scattering matrix of the matched network.

Given

$$- \Gamma_L = 0.330\angle 66.0^\circ, Z_0 = 75\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 79.52 + j53.80\ \Omega.$$

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 79.52 + j53.80$ over $75 = 1.06 + j0.72$.
- Plot it where the $r = 1.06$ circle cuts the $x = +0.72$ arc; it reads 0.158λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.330$ on refl. coeff. E or I, $S = 1.99$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.330\angle 66.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 1.99$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.65 - j0.44$, at 0.408λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.65 - j0.44$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.647$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.647$ circle.

- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.65 - j1.94$, so stub 1 must add $b_1 = -1.936 - (-0.438) = -1.498$.
 - **Solution B:** $y'_a = 0.65 - j0.06$, so $b'_1 = +0.373$.
- Read ℓ_1 off the rim, starting at the open-circuit point ($y = 0$, the left of the chart): $\ell_1 = 0.344\lambda$ (A) and 0.057λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.45$ (A), $y'_2 = 1.00 - j0.45$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.446$ (A) or $+0.446$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 75\Omega$. Matched.
- Off the rim from the open-circuit point ($y = 0$, the left of the chart): $\ell_2 = 0.312\lambda$ (A) and 0.067λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.65 - j0.44$	$0.65 - j0.44$
Stub 1 susceptance b_1	-1.498	$+0.373$
Stub 1 length ℓ_1	0.344λ	0.057λ
y after the spacing	$1.00 + j2.45$	$1.00 - j0.45$
Stub 2 susceptance b_2	-2.446	$+0.446$
Stub 2 length ℓ_2	0.312λ	0.067λ

Both are correct.

Quote either, but say which you chose. Solution B uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

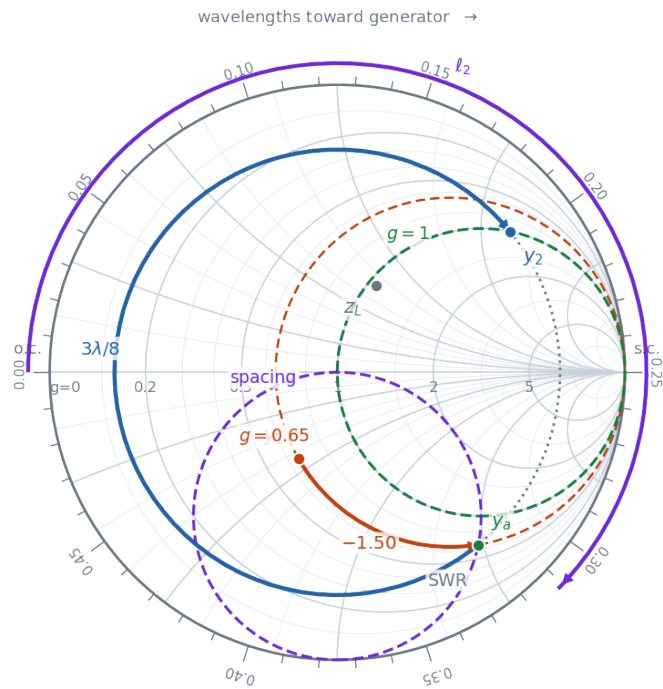
- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.330\angle 66.0^\circ]$, with $S = 1.99$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.

Microstrip realisation

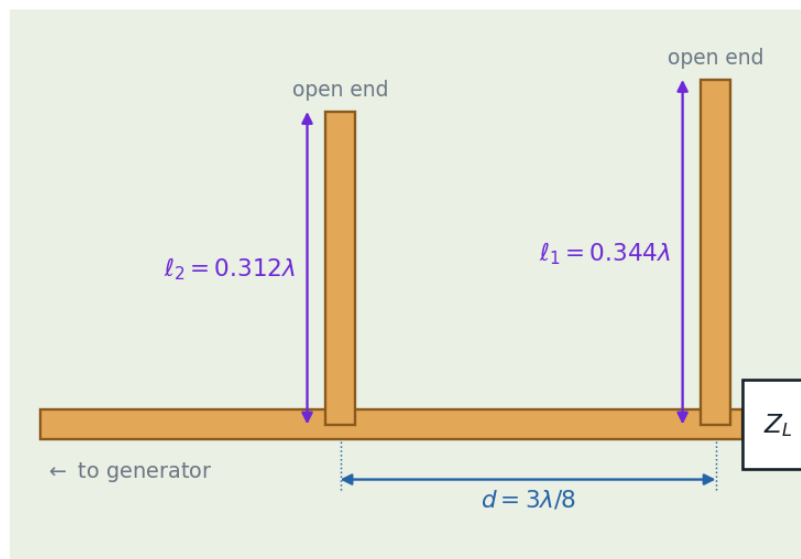
- Take FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm. Hammerstad synthesis for 75Ω gives $W/h = 0.89$, i.e. $W = 1.43$ mm, with $\epsilon_e = 3.15$.
- No frequency is given, so leave the answers in wavelengths; multiply by $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$ once f is fixed.
- Draw the stubs as T-junctions off the W -wide main line. Open stubs are plain truncated strips (no via needed) – which is why open stubs are preferred in microstrip.
- Shorten each open stub by the fringing allowance $\Delta\ell \approx 0.3h = 0.48$ mm.



solution A: $l_1 = 0.344\lambda$, $l_2 = 0.312\lambda$, open stubs $3\lambda/8$ apart

Problem 19 — Smith chart construction. Arcs are numbered in the order of the steps above.

Double stub, open-circuited, solution A



FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, 75Ω strip $W = 1.43$ mm, $\epsilon_e = 3.15$
 lengths in λ_g and to scale; strip width not to scale
 cut each open end $\Delta l = 0.48$ mm short for fringing

Problem 19 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

Problem 20 — Double-stub shunt tuner 10 2 2

82 Ba · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 82 Ba [10+2+2]

An antenna at 10 GHz presents $\Gamma = |0.45|\angle 60^\circ$ on a $100\ \Omega$ line. Design a double-stub tuner, sketch the physical diagram using microstrips, and give the S-matrices before and after matching.

Given

- $\Gamma_L = 0.450\angle 60.0^\circ$, $Z_0 = 100\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 105.98 + j103.58\ \Omega$.
- $f = 10\ \text{GHz} \Rightarrow \lambda_0 = c/f = 30.0\ \text{mm}$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 105.98 + j103.58$ over 100 = $1.06 + j1.04$.
- Plot it where the $r = 1.06$ circle cuts the $x = +1.04$ arc; it reads 0.167λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.450$ on *refl. coeff. E or I*, $S = 2.64$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.450\angle 60.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.64$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.48 - j0.47$, at 0.417λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.48 - j0.47$.

Step 4 — draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.483$. ✓

Step 5 — stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.483$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.48 - j1.86$, so stub 1 must add $b_1 = -1.856 - (-0.472) = -1.384$.
 - **Solution B:** $y'_a = 0.48 - j0.14$, so $b'_1 = +0.327$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.100\lambda$ (A) and 0.300λ (B).

Step 6 — the $3\lambda/8$ of line between the stubs

- Swing a *new* SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.77$ (A), $y'_2 = 1.00 - j0.77$ (B).

Step 7 — stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.773$ (A) or $+0.773$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 100\ \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.055\lambda$ (A) and 0.355λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.48 - j0.47$	$0.48 - j0.47$
Stub 1 susceptance	-1.384	$+0.327$
b_1		
Stub 1 length ℓ_1	0.100λ	0.300λ
y after the spacing	$1.00 + j2.77$	$1.00 - j0.77$
Stub 2 susceptance	-2.773	$+0.773$
b_2		
Stub 2 length ℓ_2	0.055λ	0.355λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

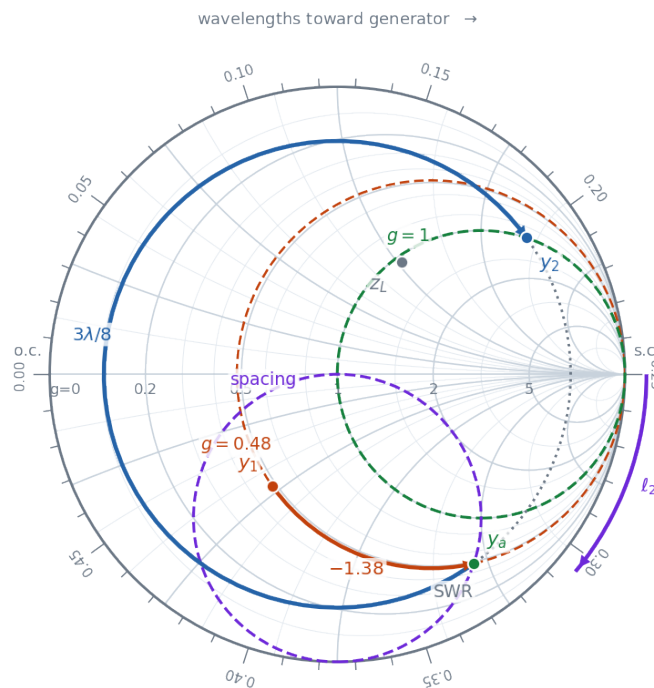
- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.450\angle 60.0^\circ]$, with $S = 2.64$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.

Microstrip realisation

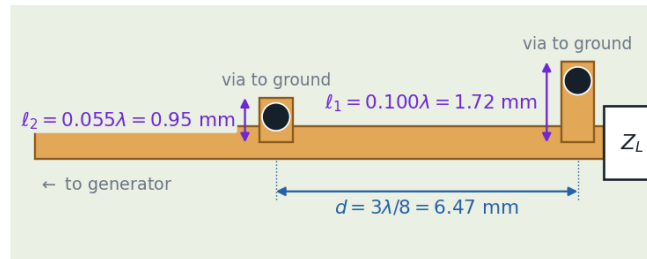
- Take FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm. Hammerstad synthesis for $100\ \Omega$ gives $W/h = 0.44$, i.e. $W = 0.71$ mm, with $\epsilon_e = 3.02$.
- $\lambda_g = \lambda_0/\sqrt{\epsilon_e} = 17.2$ mm at 10 GHz, so:
 - $\ell_1 = 0.100\lambda = 1.72$ mm.
 - $\ell_2 = 0.055\lambda = 0.95$ mm.
 - $d = 0.375\lambda = 6.47$ mm.
- Draw the stubs as T-junctions off the W -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance $\Delta\ell \approx 0.3h = 0.48$ mm.



solution A: $\ell_1 = 0.100\lambda$, $\ell_2 = 0.055\lambda$, short stubs $3\lambda/8$ apart

Problem 20 — Smith chart construction. Arcs are numbered in the order of the steps above.

Double stub, short-circuited, solution A



FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, $100\ \Omega$ strip $W = 0.71$ mm, $\epsilon_e = 3.02$, $\lambda_g = 17.2$ mm at 10 GHz; to scale

Problem 20 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

Problem 21 — Double-stub shunt tuner 12

80 Ch · $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 80 Ch [12]

Design a double-stub tuner ($3\lambda/8$ spacing) for $Z_L = 110 + j110\ \Omega$ on a $100\ \Omega$ line: find the stub lengths and the spacing.

Given

– $Z_L = 110.00 + j110.00\ \Omega$, $Z_0 = 100\ \Omega$.

Step 1 — normalise and plot z_L

- $z_L = Z_L/Z_0 = 110.00 + j110.00$ over $100 = 1.10 + j1.10$.
- Plot it where the $r = 1.10$ circle cuts the $x = +1.10$ arc; it reads 0.171λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.466$ on *refl. coeff.* *E* or *I*, $S = 2.74$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.466\angle 57.2^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.74$.

Step 2 — convert to admittance

- Draw the diameter through z_L and read the opposite intersection — a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.45 - j0.45$, at 0.421λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.45 - j0.45$.

Step 4 — draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.455$. ✓

Step 5 — stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.455$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.45 - j1.84$, so stub 1 must add $b_1 = -1.838 - (-0.455) = -1.384$.
 - **Solution B:** $y'_a = 0.45 - j0.16$, so $b'_1 = +0.293$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.100\lambda$ (A) and 0.295λ (B).

Step 6 — the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.84$ (A), $y'_2 = 1.00 - j0.84$ (B).

Step 7 — stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.844$ (A) or $+0.844$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 100\ \Omega$. Matched.

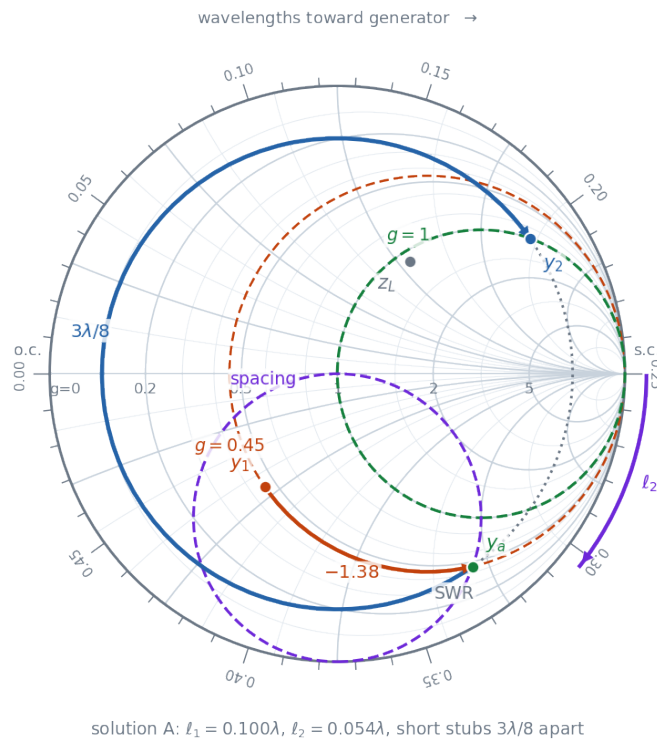
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.054\lambda$ (A) and 0.362λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.45 - j0.45$	$0.45 - j0.45$
Stub 1 susceptance	-1.384	$+0.293$
b_1		
Stub 1 length ℓ_1	0.100λ	0.295λ
y after the spacing	$1.00 + j2.84$	$1.00 - j0.84$
Stub 2 susceptance	-2.844	$+0.844$
b_2		
Stub 2 length ℓ_2	0.054λ	0.362λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 21 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 22 – Double-stub shunt tuner 4 6

78 Ch · $Z_0 = 50\Omega$

QUESTION AS PRINTED 78 Ch [4+6]

Explain the different impedance matching techniques and give the resulting S-matrix of a perfectly matched double-stub network.

Given

- $Z_L = 100.00 + j50.00\Omega$, $Z_0 = 50\Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 100.00 + j50.00$ over $50 = 2.00 + j1.00$.
- Plot it where the $r = 2.00$ circle cuts the $x = +1.00$ arc; it reads 0.213λ on the WTG scale.

- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.447$ on *refl. coeff. E or I*, $S = 2.62$ on *SWR*. Check: $(z_L - 1)/(z_L + 1) = 0.447 \angle 26.6^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.62$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.40 - j0.20$, at 0.463λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.40 - j0.20$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.400$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own* $g = 0.400$ circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - Solution A:** $y_a = 0.40 - j1.80$, so stub 1 must add $b_1 = -1.800 - (-0.200) = -1.600$.
 - Solution B:** $y'_a = 0.40 - j0.20$, so $b'_1 = +0.000$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.089\lambda$ (A) and 0.250λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j3.00$ (A), $y'_2 = 1.00 - j1.00$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -3.000$ (A) or $+1.000$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.051\lambda$ (A) and 0.375λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.40 - j0.20$	$0.40 - j0.20$
Stub 1 susceptance b_1	-1.600	$+0.000$
Stub 1 length ℓ_1	0.089λ	0.250λ
y after the spacing	$1.00 + j3.00$	$1.00 - j1.00$
Stub 2 susceptance b_2	-3.000	$+1.000$
Stub 2 length ℓ_2	0.051λ	0.375λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

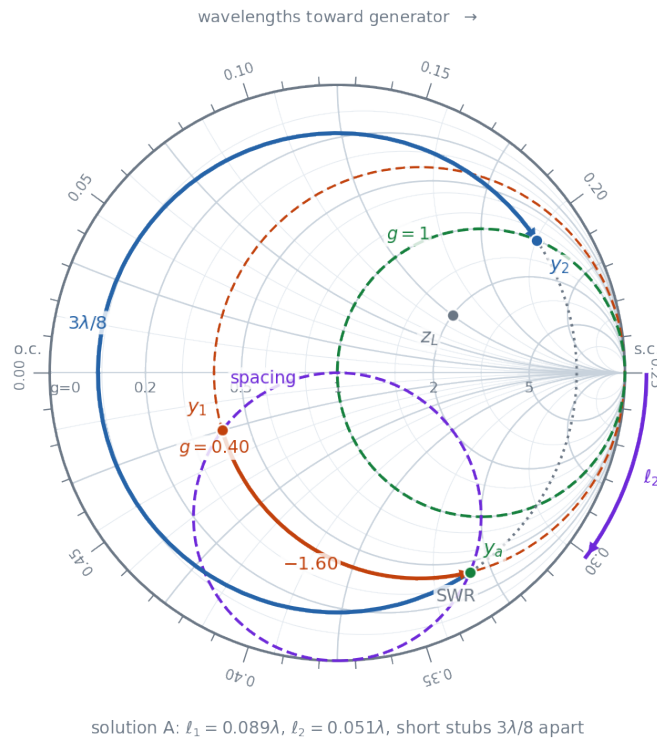
- Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.447 \angle 26.6^\circ]$, with $S = 2.62$.
- After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.

Theory half

The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



Problem 22 – Smith chart construction. Arcs are numbered in the order of the steps above.

Problem 23 – Double-stub shunt tuner 2 8

77 Ch · $Z_0 = 50 \Omega$

QUESTION AS PRINTED 77 Ch [2+8]

Sketch a double-stub perfectly matched network using microstrip and prepare its S-matrix.

Given

- $Z_L = 25.00 - j40.00 \Omega$, $Z_0 = 50 \Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 25.00 - j40.00$ over $50 = 0.50 - j0.80$.
- Plot it where the $r = 0.50$ circle cuts the $x = -0.80$ arc; it reads 0.380λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.555$ on *refl. coeff.* *E* or *I*, $S = 3.49$ on *SWR*. **Check:** $(z_L - 1)/(z_L + 1) = 0.555 \angle -93.9^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.49$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.56 + j0.90$, at 0.130λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.56 + j0.90$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto* $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.562$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own $g = 0.562$ circle*.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.56 - j1.90$, so stub 1 must add $b_1 = -1.899 - (0.899) = -2.798$.
 - **Solution B:** $y'_a = 0.56 - j0.10$, so $b'_1 = -1.000$.
- Read ℓ_1 off the rim, starting at the open-circuit point ($y = 0$, the left of the chart): $\ell_1 = 0.305\lambda$ (A) and 0.375λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.60$ (A), $y'_2 = 1.00 - j0.60$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.600$ (A) or $+0.600$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50\Omega$. Matched.
- Off the rim from the open-circuit point ($y = 0$, the left of the chart): $\ell_2 = 0.308\lambda$ (A) and 0.086λ (B).

Answer

Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.56 + j0.90$	$0.56 + j0.90$
Stub 1 susceptance	-2.798	-1.000
b_1		
Stub 1 length ℓ_1	0.305λ	0.375λ
y after the spacing	$1.00 + j2.60$	$1.00 - j0.60$
Stub 2 susceptance	-2.600	$+0.600$
b_2		
Stub 2 length ℓ_2	0.308λ	0.086λ

Both are correct.

Quote either, but say which you chose. Solution B uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

S-matrix before and after matching

- **Before** – the bare load is a one-port, so $[S] = [\Gamma_L] = [0.555\angle -93.9^\circ]$, with $S = 3.49$.
- **After** – the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$. The network is passive and reciprocal ($S_{12} = S_{21}$) and lossless, so unitarity forces $|S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

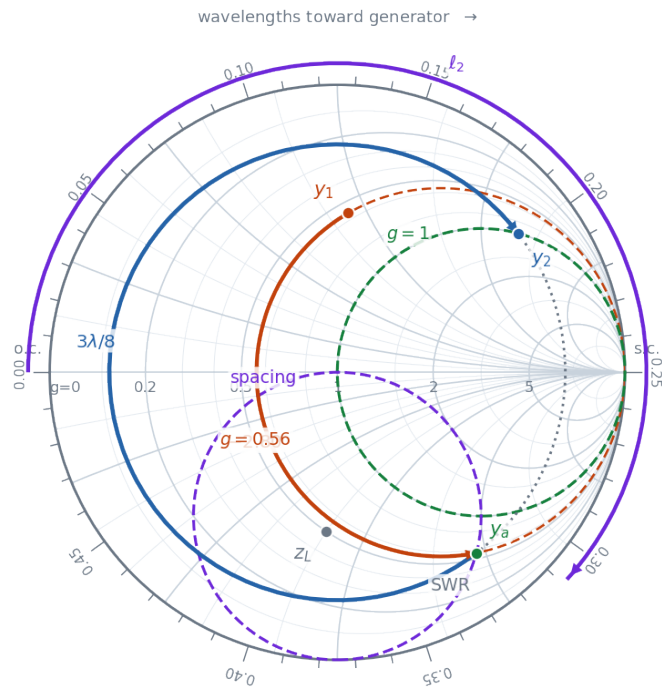
- Keeping the $e^{-j\theta}$ records the through-path electrical length; either form earns the marks if you say which you mean.

Microstrip realisation

- Take FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm. Hammerstad synthesis for 50Ω gives $W/h = 1.91$, i.e. $W = 3.06$ mm, with $\epsilon_e = 3.33$.
- No frequency is given, so leave the answers in wavelengths; multiply by $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$ once f is fixed.
- Draw the stubs as T-junctions off the W -wide main line. Open stubs are plain truncated strips (no via needed) – which is why open stubs are preferred in microstrip.
- Shorten each open stub by the fringing allowance $\Delta\ell \approx 0.3h = 0.48$ mm.

Theory half

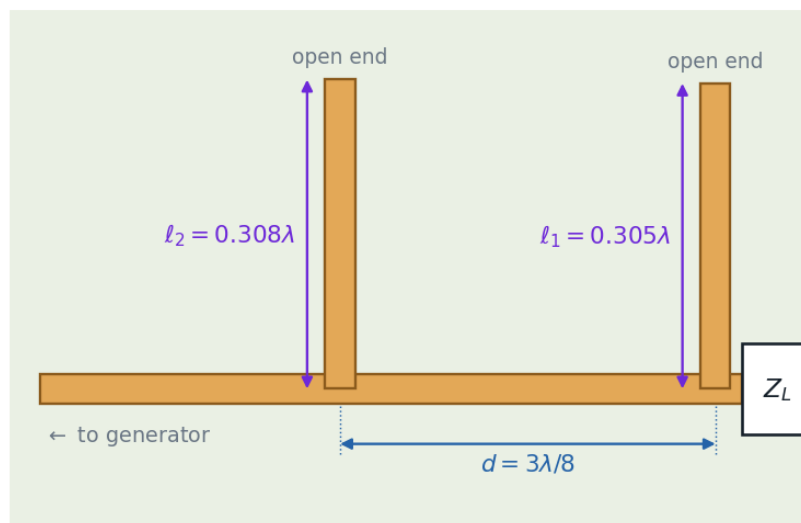
The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



solution A: $l_1 = 0.305\lambda$, $l_2 = 0.308\lambda$, open stubs $3\lambda/8$ apart

Problem 23 — Smith chart construction. Arcs are numbered in the order of the steps above.

Double stub, open-circuited, solution A



FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, 50Ω strip $W = 3.06$ mm, $\epsilon_e = 3.33$
 lengths in λ_g and to scale; strip width not to scale
 cut each open end $\Delta l = 0.48$ mm short for fringing

Problem 23 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

Problem 24 – Double-stub shunt tuner 2 8

74 Ma · $Z_0 = 50 \Omega$

QUESTION AS PRINTED 74 Ma [2+8]

State the advantages of double-stub matching over single-stub matching, and give the steps for matching a load with a double-stub network with an example, using the provided Smith chart.

Given

- $Z_L = 30.00 + j40.00 \Omega$, $Z_0 = 50 \Omega$.

Step 1 – normalise and plot z_L

- $z_L = Z_L/Z_0 = 30.00 + j40.00$ over 50 = $0.60 + j0.80$.
- Plot it where the $r = 0.60$ circle cuts the $x = +0.80$ arc; it reads 0.125λ on the WTG scale.
- Dividers from centre to z_L (the SWR-circle radius), laid on the radially scaled scales below the chart: $|\Gamma_L| = 0.500$ on refl. coeff. E or I , $S = 3.00$ on SWR. Check: $(z_L - 1)/(z_L + 1) = 0.500 \angle 90.0^\circ$, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.00$.

Step 2 – convert to admittance

- Draw the diameter through z_L and read the opposite intersection – a $\lambda/4$ half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.60 - j0.80$, at 0.375λ on the WTG scale (exactly 0.25λ from z_L).

Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and $y_1 = y_L = 0.60 - j0.80$.

Step 4 – draw the spacing circle

- Rotate the $g = 1$ circle $3\lambda/8$ toward the load (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried onto $g = 1$ by the $3\lambda/8$ of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs $g_1 \leq 1/\sin^2 \beta d = 2.00$, and here $g_1 = 0.600$. ✓

Step 5 – stub 1: slide along the constant- g circle

- A shunt stub adds susceptance only, so the point can move *only along its own $g = 0.600$ circle*.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
 - **Solution A:** $y_a = 0.60 - j1.92$, so stub 1 must add $b_1 = -1.917 - (-0.800) = -1.117$.
 - **Solution B:** $y'_a = 0.60 - j0.08$, so $b'_1 = +0.717$.
- Read ℓ_1 off the rim, starting at the short-circuit point ($y = \infty$, the right of the chart): $\ell_1 = 0.116\lambda$ (A) and 0.349λ (B).

Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through y_a and rotate **clockwise** by $3\lambda/8$.
- By construction it lands on the $g = 1$ circle: $y_2 = 1.00 + j2.53$ (A), $y'_2 = 1.00 - j0.53$ (B).

Step 7 – stub 2 cancels what is left

- Stub 2 supplies $b_2 = -2.528$ (A) or $+0.528$ (B), leaving $y_{in} = 1 + j0$, i.e. $Z_{in} = Z_0 = 50 \Omega$. Matched.
- Off the rim from the short-circuit point ($y = \infty$, the right of the chart): $\ell_2 = 0.060\lambda$ (A) and 0.327λ (B).

Answer

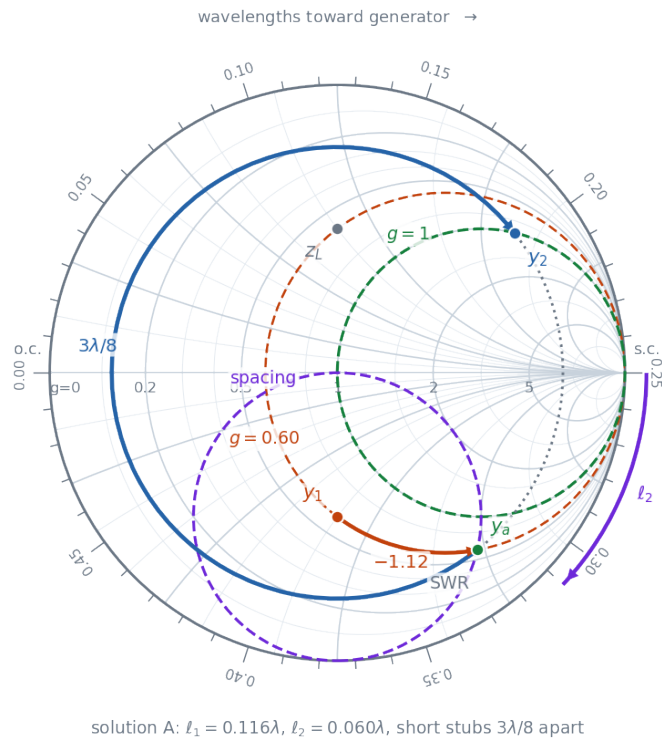
Quantity	Solution A	Solution B
Stub spacing d	$3\lambda/8$	$3\lambda/8$
First stub from load	0.000λ	0.000λ
y before stub 1	$0.60 - j0.80$	$0.60 - j0.80$
Stub 1 susceptance	-1.117	$+0.717$
b_1		
Stub 1 length ℓ_1	0.116λ	0.349λ
y after the spacing	$1.00 + j2.53$	$1.00 - j0.53$
Stub 2 susceptance	-2.528	$+0.528$
b_2		
Stub 2 length ℓ_2	0.060λ	0.327λ

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

Theory half

The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



Problem 24 — Smith chart construction. Arcs are numbered in the order of the steps above.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.